After attending the department-hosted event *Systems Savvy: A Woman's Guide to a Career in Computer Systems/Architecture*, I developed a strong interest in computer systems, particularly in improving performance and efficiency through system-level design. Presentations from leading researchers exposed me to the wide-ranging impact of systems work across domains, inspiring me to explore how modern computing infrastructure can be adapted to support increasingly complex and resource-intensive workloads. Motivated by this experience, I am excited to contribute to research that deepens my understanding of system design and its role in enabling high-performance computing.

This interest led me to pursue an ongoing research project under the supervision of Professor Qizhen Zhang, where I investigate the communication in database systems and leverage the potential to offload certain processes to dedicated hardware accelerators for greater efficiency. Specifically, I annotate the PostgreSQL source code to trace client-server interaction pathways and categorize internal functions by semantics to support the design of a cleaner abstraction layer for protocol-level communication. While this experience provides an understanding of network communication, I also expanded my knowledge in the storage engine through CSC443 lecture recordings offered by Professor Niv Dayan. Together, these experiences have given me a comprehensive understanding of database management system architecture. I can apply this knowledge to tackle systems-level challenges focused on optimizing low-level components for fast and scalable data processing, such as those explored in the Orca Lab for OracDB.

My interest in pushing system performance to its limit also extends to the field of deep learning, where computation resources have become a bottleneck for performance. To tackle these challenges, I am eager to explore how various models interact with hardware resources and develop new strategies to alleviate their memory burdens. Specifically, I am interested in both low-level approaches, such as compiler optimization across diverse hardware domains, and high-level techniques, such as quantization and pruning. To better understand potential directions for leveraging GPUs in acceleration, I participate in a GPU-focused reading group called GPU Mode. This collaborative platform explores both foundational and advanced topics in GPU architecture and its applications. Through presentations from researchers and industry professionals, I have gained insights into parallel execution, memory hierarchies, and optimization strategies, all of which have shaped my systems-oriented thinking. These experiences guide my pursuit of system-aware solutions for accelerating deep learning models in the research projects led individually by Professor Maryam Mehri Dehnavi and Professor Nandita Vijaykumar.

I look forward to contributing to research focused on computer systems and applying my knowledge to accelerate real-world applications. Thank you for considering my application.